AN IRREDUCIBLE POLYNOMIAL IS SEPARABLE EXACTLY WHEN IT HAS PAIRWISE DISTINCT ROOTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The chapter's working definition of a separable polynomial is coprimality with the derivative, which is what the final citation converts into the distinct-roots form used throughout the separability theory.

Editorial note. The label lemma-recognize-separable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-relatively-prime-polynomials (two polynomials over a field are relatively prime exactly when they share no root in an algebraic closure); separability of a polynomial as coprimality with its derivative

Source reference. Original reference 09GY: sources/stacks-fields/pinned-original-source.tex, source lines 1322–1322. Prerequisite pointer invoked at 1438; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let F be a field. An irreducible polynomial P over F is separable if and only if P has pairwise distinct roots in an algebraic closure of F.*

Proof. Suppose that $\alpha \in \overline{F}$ is a root of both P and P'. Then $P = (x - \alpha)Q$ for some polynomial Q. Taking derivatives we obtain $P' = Q + (x - \alpha)Q'$. Thus α is a root of Q. Hence we see that if P and P' have a common root, then P does not have pairwise distinct roots. Conversely, if P has a repeated root, i.e., $(x - \alpha)^2$ divides P, then α is a root of both P and P'. Combined with Lemma 09GY this proves the lemma. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.